Sequences, Series & Complex Numbers

ANSWER KEY

Section 1: Arithmetic & Geometric Progressions

1 In an arithmetic progression, the sum of the first n terms is given by $S_n = 3n^2 + 2n$. What is the 10th term a_{10}?

a. 59 ✓

b. 62

c. 65

d. 320

$a_{10} = S_{10} - S_9 = [3(100) + 20] - [3(81) + 18] = 320 - [243 + 18] = 320 - 261 = 59$.

2 If the second term of a geometric sequence is 12 and the fifth term is 96, what is the first term?

a. 3

b. 6 ✓

c. 8

d. 12

$\frac{a_5}{a_2} = r^3 = \frac{96}{12} = 8 \Rightarrow r = 2$. Then $a_1 = \frac{a_2}{r} = \frac{12}{2} = 6$.

3 Evaluate the sum of the convergent geometric series: $\sum_{n=0}^{\infty} 5(-\frac{1}{3})^n$.

a. $\frac{15}{4}$ ✓

b. $\frac{15}{2}$

c. $\frac{5}{4}$

d. 5

First term $a = 5(-1/3)^0 = 5$, common ratio $r = -\frac{1}{3}$. Sum $= \frac{a}{1-r} = \frac{5}{1-(-1/3)} = \frac{5}{4/3} = \frac{15}{4}$.

Section 2: Binomial Expansions & General Terms

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4 What is the constant term in the expansion of $(x^2 + \frac{1}{x})^9$?

a. 84 ✓

b. 36

c. 126

d. 9

General term: $\binom{9}{k}(x^2)^{9-k}(x^{-1})^k = \binom{9}{k}x^{18-3k}$. For constant term: $18 - 3k = 0 \Rightarrow k = 6$. Coefficient is $\binom{9}{6} = \binom{9}{3} = \frac{9 \times 8 \times 7}{3 \times 2 \times 1} = 84$.

5 What is the coefficient of x^4 in the expansion of $(1 - 2x)^6$?

a. 240 ✓

b. -240

c. 480

d. -480

Term with x^4: $\binom{6}{4}(1)^2(-2x)^4 = 15 \times 16x^4 = 240x^4$.

6 What is the sum of the coefficients in the expansion of $(5x - 4)^7$?

a. 1 ✓

b. 128

c. 2187

d. 0

Substitute $x = 1$: $(5(1) - 4)^7 = 1^7 = 1$.

Section 3: Operations in the Complex Plane

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7 What is the complex conjugate of $z = \frac{3-4i}{1+2i}$?

a. $-1+2i$ ✓

b. $-1-2i$

c. $1+2i$

d. $1-2i$

Multiply by $(1-2i)$: $\frac{(3-4i)(1-2i)}{1-(2i)^2} = \frac{3-6i-4i+8i^2}{1-(-4)} = \frac{-5-10i}{5} = -1-2i$. The conjugate is $-1+2i$.

8 What is the modulus of the complex number $z = (3+4i)(5-12i)$?

a. 65 ✓

b. 17

c. 13

d. 25

$|z| = |3+4i| \times |5-12i| = \sqrt{3^2+4^2} \times \sqrt{5^2+(-12)^2} = 5 \times 13 = 65$.

9 What is the polar form of $z = -3i$?

a. $3(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2})$ ✓

b. $3(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$

c. $3(\cos\pi + i\sin\pi)$

d. $-3(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$

$r = 3$. Point lies on negative imaginary axis: $\theta = \frac{3\pi}{2}$.

Section 4: De Moivre's Formula & Roots

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10 Evaluate $(\cos 30^\circ + i\sin 30^\circ)^6$.

a. -1 ✓

b. 1

c. i

d. $-i$

By De Moivre's: $\cos(6 \times 30^\circ) + i\sin(6 \times 30^\circ) = \cos 180^\circ + i\sin 180^\circ = -1 + 0i = -1$.

11 Find the real part of $(1 - i\sqrt{3})^5$.

a. 16 ✓

b. -16

c. 32

d. -32

$r = \sqrt{1+3} = 2, \theta = -\frac{\pi}{3}$. $(2e^{-i\pi/3})^5 = 32e^{-i5\pi/3} = 32(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}) = 32(\frac{1}{2} + i\frac{\sqrt{3}}{2}) = 16 + 16i\sqrt{3}$. Real part is 16.

12 How many roots of the polynomial equation $z^4 + 16 = 0$ lie in the first quadrant of the complex plane?

a. 1 ✓

b. 2

c. 3

d. 4

Roots of $z^4 = -16 = 16e^{i\pi}$ have angles $\theta_k = \frac{\pi + 2k\pi}{4} = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$. Exactly 1 angle ($\frac{\pi}{4}$) is in Quadrant I.

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